NEPSE Forecasting Study — Excluding the Thin-Floorsheet Sessions

This report records a change to the modelling design: the sessions in which the floorsheet is largely missing are no longer used to train or evaluate any model. It states the rule, what it removes, and what it does to every headline result, measured against the earlier run on the same untouched final test window. Both runs use the same scikit-learn stand-in models (CatBoost, LightGBM, XGBoost, the Transformer and the TFT could not be installed here), the same 60 predictors and the same code; the only difference is which sessions enter the panel.

Summary

Excluding the gap does not improve accuracy. On the untouched final test it costs a little at every horizon, and a lot at 10 sessions. The 1-session result survives: 58.6% against a 55.3% majority baseline (95% interval 56.5–60.6%), with the Diebold–Mariano test against climatology slightly stronger than before (Holm $p = 1.2 \times 10^{-7}$ versus 3.5×10^{-7}).

That is the useful finding. The 1-session skill is not an artefact of the degraded period: it holds after 15% of the sample, including the whole stretch where broker features are missing, is taken away. At 3, 5 and 10 sessions nothing is distinguishable from the baseline in either run, so nothing is lost there that was real to begin with.

The 10-session result degrades materially, from 55.3% to 50.5% against a 56.8% baseline. Its AUC is unchanged (0.584 → 0.588), so the ranking information is still there; what fails is calibration (expected calibration error 7.3% → 10.0%), which moves the 0.5 cut-off to the wrong place. With 15% fewer sessions the 10-session labels overlap more heavily within a smaller effective sample, and the calibration layer has less to learn from.

The backtest is unchanged in character. No strategy earns a net return distinguishable from zero after costs in either run, and the ranker's top-10 portfolio still loses money at every horizon.

The exclusion rule

The rule is measured, not assumed. For each session, coverage is the share of that session's eligible stocks that appear in the floorsheet. The distribution is sharply bimodal — 1,577 of 1,855 sessions above 95% and 261 below 30%, with almost nothing between — so the threshold is not a judgement call.

A session is **thin** when coverage is below 50%. Thin sessions that form a run of at least 5 are treated as a structural gap and dropped from the modelling panel, and so are the 19 sessions that follow each run. Isolated thin days elsewhere are kept: one missing observation inside a 20-session window is noise, not a gap.

Why the buffer matters. `amihud20`, `turnover_trend`, `trades_ratio20` and `buyer_hhi_chg5` are rolling windows over floorsheet turnover and trade counts, and the sequence model reads a 20-session

window of floorsheet columns. Without the buffer, a row in April 2023 would sit outside the gap but still read gap sessions through its own window. With it, no model input is computed from, or reads, a thin session.

Block	With buffer	Sessions	What happens in it
2022-02-10 → 2022-02-28	to 2022-03-31	13	Floorsheet coverage falls to about 43% for the month
2022-03-31 → 2023-03-30	to 2023-04-26	238	The long outage: 9–14% coverage, 16–26 stocks a session
2024-03-17 → 2024-03-31	to 2024-05-02	10	Coverage falls to about 54% for the second half of the month

277 sessions are thin in total; 317 are excluded once the blocks and their buffers are counted. The rule is implemented in the panel builder and is switched on with `NEPSE_EXCLUDE_GAP=1`, so both arms come from one code path.

What it removes

	With the gap	Without the gap
Eligible stock-sessions	360,943	305,474
Sessions	1,855	1,538
Symbols	362	362
Floorsheet coverage	88.5%	99.2%
Coverage in 2022	29.2%	99.5%
Coverage in 2023	81.2%	99.8%
1-session labelled rows	340,322	286,914
1-session up-rate	45.5%	46.2%

The excluded rows are 15.4% of the sample. Coverage across the surviving training data rises from 88.5% to 99.2%, and the up-rate rises slightly at every horizon: all three excluded blocks were falling markets — the NEPSE index lost 6.8%, 25.0% and 2.5% across them, against +114% over the study as a whole — so removing them removes a disproportionate share of “down” outcomes.

The final test window does not move. It is the last 250 sessions, 2025-10-12 to 2026-11-05, 60,526 scored rows at 1 session in both runs — the gap ends in May 2024, well before the test begins. Every comparison below is therefore on identical test rows.

Walk-forward folds after the exclusion

The folds are rebuilt over the surviving sessions, so the eight validation blocks are 91 sessions each instead of about 100. Folds 3 and 4 of the earlier run fell almost entirely inside the gap; they no longer exist in that form.

Fold	Train sessions	Validation dates	Train rows	Val rows
1	504	2021-02-28 → 2021-07-11	72,100	15,899
2	595	2021-07-12 → 2021-12-01	87,712	16,414
3	686	2021-12-02 → 2023-07-03	104,076	17,614
4	777	2023-07-04 → 2023-11-30	121,079	19,377
5	868	2023-12-03 → 2024-06-05	140,052	21,519
6	959	2024-06-06 → 2024-10-27	161,023	22,100
7	1050	2024-10-28 → 2025-03-23	183,145	22,083
8	1141	2025-03-24 → 2025-08-17	205,221	23,655

The leakage and split audit passes all ten checks on the rebuilt panel, including the truncation-recompute test: rebuilding every feature from sources truncated at 2024-10-28 reproduces the stored values exactly (max |difference| 0.0, no missing-data-pattern mismatches, 60 features × 286,755 rows).

Final-test accuracy

Both arms score the candidate their own out-of-fold log-loss selected, locked before the test was touched. At 5 sessions the no-gap run selected a stacked, Platt-scaled candidate that scored badly; the like-for-like comparison on the same `weighted` ensemble follows the table.

H	Arm	Candidate	Base	Acc	95% interval	AUC	Logloss	ECE
1d	with gap	weighted	55.3%	59.0%	57.0–60.9%	0.608	0.6720	3.23%
	no gap	weighted	55.3%	58.6%	56.5–60.6%	0.604	0.6738	3.15%
3d	with gap	weighted	57.3%	59.2%	55.7–62.6%	0.610	0.6732	4.88%
	no gap	weighted	57.3%	57.5%	54.8–60.4%	0.585	0.6795	5.69%
5d	with gap	weighted	57.3%	58.3%	53.7–62.5%	0.597	0.6801	6.23%
	no gap	stacking + platt	57.3%	57.3%	50.1–64.5%	0.473	0.6882	5.26%
10d	with gap	weighted	56.8%	55.3%	49.1–60.9%	0.584	0.6863	7.32%
	no gap	weighted	56.8%	50.5%	44.6–56.2%	0.588	0.6923	9.98%

Like for like, on the `weighted` ensemble in both arms: 1 session 59.0% → 58.6%; 3 sessions 59.2% → 57.5%; 5 sessions 58.3% → 55.2%; 10 sessions 55.3% → 50.5%. The 5-session AUC of 0.473 in the table above belongs to the selected stacked candidate, not to the ensemble; the `weighted` ensemble gives 0.578 there. That a selection rule validated out of fold can pick a candidate this poor is itself worth reporting.

The intervals are wide and overlap at every horizon, so the 1-, 3- and 5-session differences are within sampling error. The 10-session drop is not: 50.5% sits below its own 56.8% baseline, and the bootstrap probability that it beats that baseline is 0.013.

Significance and the ranker

Horizon	Arm	p (acc ≤ base)	DM vs climatology, Holm p	Rank-IC	IC-IR	Spread
1d	with gap	< 0.001	3.5e-07	0.284	2.50	1.02%
	no gap	< 0.001	1.16e-07	0.281	2.49	1.02%
3d	with gap	0.144	0.00632	0.270	2.12	1.45%
	no gap	0.418	0.0418	0.259	2.03	1.45%
5d	with gap	0.317	1.0	0.248	1.67	1.67%
	no gap	0.515	1.0	0.234	1.65	1.61%
10d	with gap	0.685	1.0	0.213	1.35	2.04%
	no gap	0.987	1.0	0.195	1.33	1.91%

The cross-sectional ranker is almost untouched: mean rank-IC 0.284 → 0.281 at 1 session and 0.213 → 0.195 at 10, with the share of sessions whose IC is positive unchanged at 98.3% and 93%. Whatever the ranker is reading, the gap period was neither feeding it nor fooling it.

Backtest

Long-only, entry at the next session's open, 10 bp slippage a side plus commission and the 7.5% short-term capital-gains tax. Mean net return per trade, with its session-block bootstrap interval:

Strategy	Arm	Trades	Mean net / trade	95% interval	P(mean ≤ 0)
1 session, p ≥ 0.55	with gap	1,409	-0.332%	-0.507 to -0.158%	1.000
	no gap	1,371	-0.401%	-0.582 to -0.228%	1.000
3 sessions, p ≥ 0.60	with	244	0.545%	-0.251 to 1.406%	0.118

Strategy	Arm	Trades	Mean net / trade	95% interval	P(mean ≤ 0)
	gap				
	no gap	239	0.446%	-0.343 to 1.501%	0.139
1 session, ranker top 10	with gap	2,392	-0.525%	-0.660 to -0.382%	1.000
	no gap	2,393	-0.516%	-0.657 to -0.381%	1.000
10 sessions, ranker top 10	with gap	2,315	-0.405%	-1.266 to 0.494%	0.815
	no gap	2,317	-0.381%	-1.395 to 0.691%	0.751

Nothing changes qualitatively. The only strategies with a positive point estimate in either run are the 3-session and 10-session p ≥ 0.60 filters, and both intervals comfortably straddle zero.

What to report

Report the no-gap run as the primary result. It is the cleaner design: no model is trained on sessions where nine stocks in ten have no broker, trade-count or turnover data, and no feature window reads one. That the 1-session result survives the change, with a slightly stronger Diebold–Mariano statistic, is a better claim than the marginally higher number obtained by training on the degraded period.

Report the with-gap run beside it as a sensitivity check, and say plainly that excluding the gap costs rather than gains accuracy. A reviewer who asks "what happens if you drop the period where your microstructure features are missing?" should find the answer already in the paper.

Do not claim the exclusion improved anything. It did not. What it bought is a claim that does not depend on a period of degraded data, and a 10-session result that is now visibly worse than its baseline rather than quietly indistinguishable from it.

Both runs still use scikit-learn stand-ins. CatBoost, LightGBM, XGBoost, the Transformer and the TFT could not be installed in this environment. If the native lite-panel run is repeated, the same exclusion should be applied there before its numbers are compared with anything here.

Reproduction

Stages 1–2 (the 91-million-trade ingest) were not rerun; the cleaned daily floorsheet from the earlier run was reused, and stage 01b onwards was rebuilt from the original SQL dumps. The rebuild was verified against the earlier run before the exclusion was switched on: panel rows 405,491, eligible 360,943, 362 symbols, 1,855 sessions and all four label counts reproduced exactly, so the two arms differ only by the rule.

`NEPSE_EXCLUDE_GAP=1 FAMILY=sklearn START=3 bash run_pipeline.sh` rebuilds the panel with the exclusion and reruns stages 3–14. The thresholds are `GAP_MIN_COVERAGE`, `GAP_MIN_BLOCK`

and `GAP_BUFFER` in `common.py`. Reports for this run are in `nepse_results_nogap.zip`; the patched source is in `nepse_pipeline_nogap.zip`.